

Primitive-free resonant response and descent obstruction

primitive-free resonant
boundary-response pair

$$\beta = \xi \smile \xi$$

$$\mathcal{L}_{\text{res}} = \mathbb{C}\beta \subset B^2$$

$$\mathcal{P}_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}})/Z^1$$

$d \circ \mathcal{P}_{\text{res}} \rightarrow \mathcal{L}_{\text{res}}$
is an isomorphism

$$T = H^1(C^\bullet) = \mathbb{C}\alpha \oplus \mathbb{C}\xi$$

$$T_{\text{resp}} = \mathbb{C}\alpha$$

$[\xi, \xi] = 2\beta \neq 0$ excludes the ξ direction

$$\Theta(a) = \bar{D}_a \bar{d}^{-1}, \quad \Theta(a) = \rho_{\text{res}}(a) \text{id}, \quad \rho_{\text{res}}(\alpha) = 2$$

maximal descent domain
and obstruction

$$Z = Z^1, \quad V_{\text{res}} = d^{-1}(\mathcal{L}_{\text{res}}), \quad \pi: C^2 \rightarrow C^2/\mathcal{L}_{\text{res}}$$

$$\mathfrak{o}: T \rightarrow \text{Hom}(Z, C^2) \oplus \text{Hom}(V_{\text{res}}, C^2/\mathcal{L}_{\text{res}})$$

$$a \mapsto (D_a|_Z, \pi D_a|_{V_{\text{res}}})$$

$$T_{\text{resp}} = \ker \mathfrak{o} = \mathbb{C}\alpha$$

For $a = s\alpha + t\xi$ with $t \neq 0$:
 $D_a \xi = [s\alpha + t\xi, \xi] = 2t\beta$
 If QD_a descends through V_{res}/Z , then $Q(\beta) = 0$.
 Output quotienting cannot rescue the ξ direction.

primitive-independent marked object = $(\mathcal{L}_{\text{res}}, T_{\text{resp}}, \rho_{\text{res}})$